

Exam Date & Time: 11-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech IV Semester - End Semester Examination - April / May -2024

Linux Programming [CSE-4303]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Illustrate the working of two major components of Linux operating system in detail. (5)
 - A)
 - B) Outline the advantages of Linux operating system. (3)
 - C) Identify the type of following commands: (2)
 - i) \$pwd
 - ii) \$chown
 - iii) \$cp
 - iv) \$cat
- 2) Solve the following using wild cards: (5)
 - A)
 1. List all the filenames whose name starts with at least 2 characters and followed by any numbers of characters.
 2. List all the filenames whose name starts with a single letter followed by a single digit and then any number of alphanumeric characters.
 3. List all the filenames with exactly only having 3 letters.
 4. List all the filenames having only alphabets.
 5. List all the filenames whose extension is '.c' and '.txt'.
 - B) Demonstrate any two options of 'ls' command. (3)
 - C) Illustrate the working of following commands: (2)
 1. mkdir
 2. cd
- 3) Construct a shell script to accept 2 strings as input and perform string concatenation. (5)
 - A)
 - B) Illustrate the meaning of following command lines. (3)
 - i) cd dir1

- ii) vi test.sh
- iii) sh test.sh

- C) Build the command line using redirection operators for the following:
i) Count the number entries in /etc/passwd. (2)
ii) Create a new file without using vi editor.
- 4) Construct a shell script to accept 'N' integers into an array and find the sum of array elements. (5)
- A)
- B) Construct the command lines using 'sed' command:
- 1. Edit the word 'Unix' by 'Linux'.
 - 2. Replace the word 'old' by 'new'. (3)
 - 3. Delete all the lines having the word 'pathname'.
- C) Illustrate the differences between shell variables and user defined variables in Linux. (2)
- 5) Develop the appropriate vi editor commands for the following:
- A)
- 1. To open a new file with '.c' extension.
 - 2. Enter in to the 'insert mode'.
 - 3. Save your work and exit to terminal. (5)
 - 4. Move the cursor 3 lines up.
 - 5. Delete the entire current line.
- B) Build the 'awk' programming lines for the following:
- 1. Print the columns 1 and 5 from a file.
 - 2. Perform the addition of the digits 5 and 8. (3)
 - 3. To check whether the value is 3rd column is greater than 100 or not.
- C) Rephrase the following command lines correctly as per their syntax:
- 1. -f -e \$ps (2)
 - 2. aux \$ps

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